Politecnico di Milano

Dipartimento di Elettronica, Informazione e Bioingegneria

prof. Fabrizio Ferrandi

Advanced Algorithms and Parallel Programming I part–February 14th, 2023

Polimi ID _____ _____

Surname _____ _____ **Name** _____

- This is a closed-book examination. You cannot use computers, phones, or laptops during the exam.
- Paper will be provided, but you should bring and use writing instruments that yield marks dark enough to be read easily. Erasable pens can be used.
- Total available time: 1h:15m.

Exercise 1 (4 points) _____

Exercise 2 (4 points) _____

Exercise 3 (4 points) _____

Exercise 4 (4 points) _____

Exercise n. 1

Answer the following questions about parallel application modeling and briefly explain (without an explanation, the answer will be considered invalid)

A. Please define a PRAM. (1)

B. Please describe what "Common CRCW" means. (1)

C. Please define the speed-up of a parallel algorithm according to the PRAM model (1)

D. Please describe how Gustafson defines the speed-up. (1)

Exercise n. 2

Answer the following questions about algorithms and briefly explain (without an explanation, the answer will be considered invalid)

A. Please describe how the Divide and Conquer class of algorithms works. (2)

B. How the complexity of a Divide and Conquer algorithm can be computed? (1)

C. Please discuss the complexity of the "Powering a number" algorithm. (1)

Exercise n. 3

Answer the following questions about the advanced algorithm and briefly explain (without an explanation, the answer will be considered invalid)

A. Please explain what the memoization technique is meant for and where it matters. (2)

B. Describe the skip-list data structure, how the search is performed, and which complexity has. (2)

Exercise n. 4

Consider a dynamic table data structure for storing a variable number of elements in memory.
When created, the table has space for storing only one element, and pushing one element in the table has a constant cost.

If the table is full, twice the size of the original table is allocated to a new table and all elements of the original table are copied in the new table. The cost for allocating one unit of memory is constant.

Compute the real cost for inserting n elements into a new-allocated table (having an initial capacity for storing only one element) and determine the amortized cost for the allocating the same number of elements using one of the methods seen during the course.